

Short-Term Solar Irradiance Forecasting in Tropical Andean Climates: A Graph Neural Network Approach with Conformal Uncertainty Quantification

Esteban Ladino Nieto^a

^a*Department of Industrial Engineering, Universidad de los Andes, Bogotá, Colombia*

Abstract

Solar photovoltaic (PV) generation is expanding rapidly across tropical Latin America: Colombia alone has surpassed 1 GW of installed capacity and targets 9 GW by 2030 under its National Energy Plan [1]. Reliable short-term forecasts of global horizontal irradiance (GHI) are a prerequisite for efficient operation of PV-integrated power systems, supporting reserve scheduling, reducing imbalance penalties, and enabling real-time dispatch decisions. In tropical countries such as Colombia, the forecasting problem is compounded by the high frequency and spatial heterogeneity of convective cloud systems, which can reduce surface irradiance by more than 80% within minutes and are poorly captured by coarse-resolution numerical weather prediction models. This paper proposes a graph neural network framework for short-term GHI forecasting from GOES-16 geostationary satellite imagery and couples it with conformal uncertainty quantification (UQ) to produce calibrated prediction intervals suitable for risk-aware operational decisions. We introduce **GraphSAGE-LSTM**, which treats each satellite patch as a graph of spatially connected pixels, performs neighbourhood aggregation via GraphSAGE with inverse-distance edge weights, and passes the resulting embedding sequence through an LSTM. The model is evaluated against a ResNet-LSTM baseline, a flat-MLP spatial-ablation baseline, and persistence, at two Colombian sites representing contrasting climatic regimes: El Paso (César), a high-irradiance low-elevation tropical site, and Universidad

Email address: e.ladino@uniandes.edu.co (Esteban Ladino Nieto)

de los Andes (Bogotá), a high-altitude equatorial mountain station with intense convective variability. Across forecast horizons of 1, 3, and 6 hours and four independent random seeds, GraphSAGE-LSTM achieves the best skill among the deep-learning models at the 6-hour horizon at both sites ($\text{Skill}_{\text{day}} = 0.636 \pm 0.013$ vs. 0.614 ± 0.023 for ResNet-LSTM at El Paso; 0.417 ± 0.008 vs. 0.368 ± 0.019 at Uniandes), with consistently lower seed-to-seed variance — while the markedly lower absolute skill at the high-altitude site indicates that site-specific cloud dynamics should inform model selection in operational forecasting pipelines. Split Conformal Prediction is applied as a post-hoc UQ layer, providing finite-sample marginal coverage guarantees without distributional assumptions. This work establishes a reproducible GOES-16 benchmark for tropical Andean sites — a climate regime markedly underrepresented in the solar forecasting literature — and lays the foundation for a staged UQ roadmap toward adaptive-width intervals and aleatoric/epistemic uncertainty decomposition.

Keywords: Solar irradiance forecasting, GOES-16, graph neural networks, GraphSAGE, LSTM, deep learning, conformal prediction, uncertainty quantification, clear-sky index, tropical climate, renewable energy integration

1. Introduction

Solar photovoltaic (PV) generation has grown rapidly across Latin America, with Colombia reaching over 1 GW of installed capacity and targeting 9 GW by 2030 under its National Energy Plan [1]. Reliable short-term forecasts of
5 global horizontal irradiance (GHI) are a prerequisite for efficient operation of PV-integrated power systems: they enable reserve scheduling, reduce imbalance penalties, and support real-time dispatch and grid management. The growing share of variable renewable generation makes this forecasting problem increasingly important: unlike slowly-varying demand, cloud-induced irradiance
10 variability is continuous, stochastic, and correlated across spatially distributed generation assets, and accurate, uncertainty-aware short-term forecasts provide operators with the situational awareness needed to schedule reserves, trigger

demand response, and avoid costly imbalances.

In tropical countries such as Colombia, the forecasting challenge is compounded by the high frequency and spatial heterogeneity of convective cloud systems, which can reduce surface irradiance by more than 80% within minutes and are poorly captured by coarse-resolution numerical weather prediction (NWP) models. Most published forecasting benchmarks, however, are built around temperate or semi-arid climates, leaving tropical and high-altitude equatorial sites — where cloud dynamics differ substantially — underrepresented in the literature.

Satellite-based nowcasting methods bypass NWP limitations by inferring cloud motion and optical depth directly from geostationary imagery. Instruments such as the GOES-16 Advanced Baseline Imager (ABI) provide multispectral observations every 10 minutes at sub-kilometre resolution, making them well-suited for sub-hourly to hourly irradiance estimation and forecasting. The dominant paradigm couples a spatial feature extractor (applied to a patch of satellite pixels centred on the target site) with a temporal model that exploits the recent history of observed GHI or satellite state [2, 3].

Deep learning has substantially advanced this paradigm. Convolutional neural networks (CNNs) applied to satellite patches learn translation-invariant spatial patterns associated with cloud regimes, and when combined with recurrent models such as LSTMs they achieve state-of-the-art performance on benchmark datasets [4, 5, 6]. More recently, graph neural networks (GNNs) have emerged as an alternative spatial encoder: by representing pixels as nodes in a graph and learning neighbourhood aggregation rules, GNNs relax the translation-invariance assumption of CNNs and can capture irregular spatial dependencies introduced by non-uniform cloud structures [7, 8]. However, systematic comparisons of GNN-based and CNN-based encoders for GHI forecasting remain scarce, and none, to the best of our knowledge, address the tropical Andean climate.

Contributions

This paper makes the following contributions:

1. We propose **GraphSAGE-LSTM**, a model that treats a GOES-16 satellite patch as a k -nearest-neighbour weighted pixel graph, applies GraphSAGE neighbourhood aggregation [7] with inverse-distance edge weights, and passes the resulting sequence of graph embeddings through an LSTM to produce a point forecast of GHI at horizons of 1, 3, and 6 hours.
2. We conduct a **rigorous empirical comparison** across two climatologically distinct Colombian sites, three forecast horizons, and four random seeds, including Bayesian hyperparameter optimisation via Optuna [9] for GraphSAGE-LSTM, ResNet-LSTM, and a flat MLP spatial-ablation baseline.
3. We show that GraphSAGE-LSTM **achieves the best skill among the deep-learning models** at the 6-hour horizon at both sites (El Paso: $\text{Skill}_{\text{day}} = 0.636$ vs. 0.614 for ResNet-LSTM; Uniandes: 0.417 vs. 0.368), with consistently lower seed-to-seed variance, while the absolute skill of both models is markedly lower at the climatologically harder high-altitude site — evidence that site-specific cloud dynamics, not just model architecture, govern attainable forecasting skill.
4. We implement **Split Conformal Prediction** [10, 11] as a post-hoc UQ layer applicable to any trained backbone. The calibration procedure produces marginal-coverage-valid prediction intervals without distributional assumptions, supporting risk-aware operational decisions for solar-integrated power systems. This constitutes Stage 1 of a staged UQ roadmap: Split CP (this work) $\rightarrow$ Conformalized Quantile Regression [12] for adaptive-width intervals $\rightarrow$ Bayesian uncertainty decomposition [13] for aleatoric/epistemic separation.
5. We establish a **reproducible GOES-16 benchmark for tropical Andean sites**, including data manifests, patch extraction, model code, Optuna studies, and evaluation scripts, addressing the scarcity of published

satellite-based GHI forecasting benchmarks for Latin American and equatorial tropical contexts.

Paper organisation

75 Section 2 reviews related work. Section 3 describes the datasets, model architectures, training protocol, and evaluation metrics. Section 4 presents quantitative results. Section 5 analyses key findings. Section 6 concludes and outlines future work.

2. Related Work

80 *2.1. Short-term solar irradiance forecasting*

Short-term GHI forecasting methods span a wide spectrum from statistical time-series models to physically-based NWP and data-driven machine learning [14, 15]. For intra-hour and sub-day horizons (0–6 h), satellite-based and ground-observation methods generally outperform NWP, which requires several
85 hours of spin-up time [16].

Persistence (assuming future irradiance equals the most recent observation) is the standard benchmark for short-term forecasting. It is competitive at very short horizons (under 30 minutes) when irradiance changes slowly, but degrades rapidly beyond one hour [17].

90 **Statistical models** such as ARIMA and its seasonal variant SARIMA have been applied to GHI series after removing the deterministic clear-sky envelope [18, 19]. Working on the clear-sky index $k_t = \text{GHI}/\text{GHI}_{\text{cs}}$ improves stationarity and allows standard time-series machinery at hourly resolution. Despite their simplicity, seasonal ARIMA models can be competitive at multi-hour horizons
95 in climates with regular diurnal patterns [20].

2.2. Deep learning with satellite imagery

CNN-based architectures applied to sequences of geostationary satellite patches have become the dominant approach for satellite-based GHI forecasting. Early

work showed that cloud-motion vectors extracted from visible channels improve
100 short-term forecasts [21]. More recent studies feed raw satellite reflectances di-
rectly into CNNs followed by fully-connected regressors or LSTMs [5, 6, 22].
The encoder-decoder design, where a CNN extracts per-frame spatial embed-
dings that are then processed by a recurrent model, has shown consistent gains
over purely spatial or purely temporal baselines [23].

105 Attention-based architectures including Vision Transformers (ViT) and Swin
Transformers have also been applied to satellite-based solar forecasting [24, 25],
achieving competitive performance at the cost of increased parameter count
and data requirements. For moderate dataset sizes such as those encoun-
tered in single-site studies, CNN+LSTM hybrids remain a practical and well-
110 characterised baseline.

2.3. Graph neural networks for spatial data

GraphSAGE [7] is an inductive GNN that learns aggregation functions over
local neighbourhoods, enabling generalisation to unseen nodes. It has been
applied to traffic forecasting [26], air quality prediction, and wind power gener-
115 ation [27], where spatial proximity defines the graph topology. In the context of
satellite imagery, representing pixels as graph nodes allows the model to capture
irregular spatial dependencies that periodic CNN filters may miss, such as cloud
boundaries or fractional cloud cover patterns.

Closest to our work, [8] applied a GNN-LSTM model to multi-site solar irra-
120 diance forecasting, treating monitoring stations as nodes. Our approach differs
in that we build the graph within a single satellite patch (pixel level), without
requiring multi-station measurements, which is more practical for single-site
deployment.

2.4. Forecasting in tropical and Andean climates

125 Most GHI forecasting benchmarks are conducted in temperate or semi-arid
climates (Europe, southwestern United States, Australia) [28, 29]. Tropical An-
dean sites present distinct challenges: convective cloud formation is driven by

topographic lifting and moisture advection, producing strong intraday variability concentrated in the afternoon [30]. Few published studies address satellite-based forecasting for Colombian or equivalent tropical mountain conditions, motivating the present work.

2.5. Forecasting for power system operation and decision-making

Accurate short-term irradiance forecasts feed directly into operational decision-making for solar-integrated power systems: they support reserve scheduling, unit commitment, and demand-response triggers, reducing the cost of accommodating variable renewable generation [31]. For renewable integration specifically, accurate forecasting reduces the need for spinning reserves and allows pre-emptive load balancing, directly limiting the impact of generation uncertainty on grid stability.

Beyond point forecasts, uncertainty quantification (UQ) methods that accompany predictions with calibrated confidence intervals enable operators to make risk-aware dispatch decisions: when forecasts carry explicit probabilistic guarantees, reserve scheduling can be optimised to a desired risk level rather than relying on conservative worst-case margins. Distribution-free approaches such as conformal prediction [10, 11] are particularly attractive for operational deployment because they provide finite-sample coverage guarantees without assumptions on the residual distribution — a property that is valuable in tropical climates, where forecast errors are heavy-tailed due to abrupt cloud passages.

This work contributes to this literature by (i) developing spatially-aware deep learning for solar forecasting at tropical Andean sites that remain underrepresented in existing benchmarks, and (ii) coupling point forecasters with distribution-free conformal prediction intervals suitable for integration into operational, risk-aware dispatch pipelines.

Table 1: Study site metadata.

	El Paso (César)	Uniandes (Bogotá)
Latitude	9.737°N	4.602°N
Longitude	73.695°W	74.066°W
Elevation	50 m a.s.l.	2 600 m a.s.l.
Climate	Tropical semi-arid	Equatorial mountain
Avg. GHI	~5.5 kWh/m ² /day	~4.2 kWh/m ² /day

3. Methodology

155 3.1. Study sites

Experiments are conducted at two meteorological stations in Colombia that represent climatologically contrasting regimes (Table 1).

El Paso is located in the lowland Caribbean region of César department, characterised by high annual irradiance, predominantly clear-sky conditions, and dry-season cloud cover driven by the Inter-Tropical Convergence Zone (ITCZ).
160

Universidad de los Andes (referred to as Uniandes) is an instrumented rooftop station in the Colombian Andean city of Bogotá. At 2 600 m above sea level, it experiences the equatorial bimodal rainfall regime with intense convective cloud formation typically peaking in the early afternoon, producing sharp
165 intraday GHI drops.

3.2. Data

3.2.1. Ground measurements

GHI is measured at 10-minute resolution and stored in time-zone-aware (UTC) Parquet files. Missing values account for less than 2% of the record
170 at both sites and are excluded during dataset construction.

3.2.2. Satellite data and patch extraction

We use the GOES-16 ABI Level-2 Multi-Band Cloud and Moisture Imagery Product (ABI-L2-MCMIPF), which provides all 16 ABI spectral channels at a

unified 10-minute cadence and ~ 2 km spatial resolution in the visible bands.

175 Raw hourly ABI-L2-MCMIPF granules are pre-processed into compact per-hour archives: each archive is a compressed NumPy array of shape $(6, 16, P, P)$, holding the six 10-minute slots within that hour, the $C = 16$ spectral channels, and a $P \times P = 16 \times 16$ pixel patch ($P = 16$) centred on the target site, in units consistent with each band’s native reflectance or brightness temperature.

180 At inference/training time, a single timestamp is resolved to its $(C, P, P) = (16, 16, 16)$ frame by selecting the corresponding hourly archive and 10-minute slot, so no redundant decoding of the original satellite granules is required.

3.2.3. Manifests, sequence construction, and temporal split

For the deep-learning models (ResNet-LSTM, GraphSAGE-LSTM, FlatMLP),

185 forecast samples are constructed as pairs (X, y) where the input $X \in \mathbb{R}^{L \times C \times P \times P}$ is a sequence of $L = 24$ satellite patch frames covering the most recent 4 hours of observation (10-minute cadence), and the target $y \in \mathbb{R}$ is the ground-measured GHI at the site H steps ahead ($H \in \{6, 18, 36\}$ steps, corresponding to 1, 3, and 6 hours).

190 A pre-computed **manifest** (one Parquet table per site and horizon) enumerates every valid forecast sample: for each candidate target timestamp it stores the label y , the list of $L = 24$ history timestamps **history_ts**, and a coverage flag confirming that (i) all L history steps exist in the ground-truth series and (ii) a satellite archive is available for each of them. Samples for which any of the L history frames is missing on disk are filtered out at load time, so every

195 batch the models see contains a complete 4-hour satellite history. Two thin **Dataset** wrappers consume the same manifest and patch archives but reshape the retrieved frames differently for the two model families:

- **PatchSeqDataset** (used by ResNet-LSTM and FlatMLP) returns each
- 200 sample as an image-like sequence $X \in \mathbb{R}^{L \times C \times P \times P}$ — i.e., L stacked $(16, 16, 16)$ tensors preserving the 2-D pixel grid, ready to be consumed by 2-D convolutions.

Table 2: Temporal data splits.

Site	Split	Start	End	n (approx.)
El Paso	Train	2022-03-01	2023-06-30	69 984
	Val	2023-07-01	2023-10-31	17 568
	Test	2023-11-01	2024-03-07	18 288
Uniandes	Train	2023-09-01	2024-09-30	56 850
	Val	2024-10-01	2024-12-31	13 104
	Test	2025-01-01	2025-03-28	12 384

- **GraphSeqDataset** (used by GraphSAGE-LSTM) returns each sample as a node-feature sequence $X \in \mathbb{R}^{L \times N \times C}$ with $N = P^2 = 256$: every frame is transposed and reshaped from (C, P, P) to (P^2, C) , so that each of the 256 pixels becomes a graph node carrying a $C = 16$ -dimensional feature vector (its 16-channel spectral signature at that location).

To prevent data leakage, train, validation, and test sets are defined by non-overlapping calendar blocks (Table 2), and the manifest coverage check, splitting boundaries, and patch archives are shared across all three deep-learning models, ensuring that ResNet-LSTM, GraphSAGE-LSTM, and FlatMLP are trained and evaluated on *exactly* the same samples.

3.3. Models

3.3.1. Persistence baseline

The persistence model assumes irradiance is unchanged over the forecast horizon: $\hat{y}(t+H) = \text{GHI}(t)$. It is optimal under no-change dynamics and serves as the standard benchmark for short-term solar forecasting.

3.3.2. ResNet-LSTM: convolutional spatial encoder

ResNet-LSTM is an encoder-decoder that treats each satellite frame as a conventional multi-channel image and lets a convolutional network learn its

spatial representation, before a recurrent network reasons over the sequence of representations.

Spatial encoder (per frame). Each frame $X_\ell \in \mathbb{R}^{C \times P \times P}$ ($\ell = 1, \dots, L$) is passed independently through a lightweight ResNet [32]: a 3×3 convolutional stem, followed by three residual stages of **BasicBlock** units (two 3×3 convolutions with batch normalisation and a skip connection). The first stage preserves the 16×16 spatial resolution, while the second and third stages apply stride-2 downsampling, taking the spatial grid through $16 \rightarrow 8 \rightarrow 4$. A global-average-pooling layer then collapses the 4×4 feature map to a single vector, which a final linear layer projects to a d_e -dimensional frame embedding $\mathbf{e}_\ell \in \mathbb{R}^{d_e}$ (**emb_dim**, tuned in $\{64, 128, 192, 256\}$). The encoder’s channel width is controlled by **base** $\in \{16, 24, 32, 48\}$ (the three stages use **base**, $2 \times \text{base}$, and $4 \times \text{base}$ channels respectively); the resulting parameter count ranges from approximately 50k to 500k depending on the configuration. Because the same encoder weights are applied to every one of the $L = 24$ frames, the model is translation-equivariant and shares a single spatial template across the whole 4-hour history.

Temporal encoder. The L frame embeddings are normalised (**LayerNorm**) and stacked into a sequence $(\mathbf{e}_1, \dots, \mathbf{e}_L) \in \mathbb{R}^{L \times d_e}$, which a single- or two-layer LSTM [33] (**hidden_t**, **n_lstm_layers**) processes step by step; only its final hidden state $\mathbf{h}_L \in \mathbb{R}^{d_t}$ is retained. A two-layer MLP head (**Linear** $\rightarrow$ **ReLU** $\rightarrow$ **Dropout** $\rightarrow$ **Linear**) maps $\mathbf{h}_L$ to the scalar point forecast $\hat{y}$.

3.3.3. GraphSAGE-LSTM: graph-based spatial encoder

GraphSAGE-LSTM shares the same encoder–decoder skeleton as ResNet-LSTM — spatial encoder per frame, then LSTM over the embedding sequence, then MLP head — but *replaces the convolutional encoder with a graph neural network* that treats individual pixels, rather than the whole frame, as the basic spatial unit.

From pixels to a graph. Each (C, P, P) frame is reshaped (by **GraphSeqDataset**, Section 3.2.3) into $N = P^2 = 256$ nodes, one per pixel, each carrying its native

$C = 16$ -channel spectral vector as a node feature — no convolution or pooling is applied beforehand. A single k -nearest-neighbour graph $G = (V, E, \mathbf{w})$ is built once over the 16×16 pixel grid (by Euclidean distance in 2-D pixel coordinates) and reused for every frame and every sample: each node connects to its k closest pixels ($k \in \{4, 8, 12, 16\}$, tuned per site/horizon, Section 3.4), and every edge is assigned an inverse-distance weight $w_{vu} = 1/d(v, u)$ (Eq. 1 below). Larger k values progressively add diagonal, two-step, and knight-move neighbours, so k effectively controls the spatial receptive field of a single graph layer. Unlike a fixed 8-connected grid or a convolutional filter, this k -NN topology lets the model learn *different* aggregation weights for differently-located pixels (e.g. patch centre vs. periphery), rather than sliding one shared template across the image. The edge index and weights are precomputed once, registered as model buffers, and replicated across the batch dimension, so the same mini-batch `DataLoader` and training loop used for ResNet-LSTM apply unchanged.

$$w_{vu} = \frac{1}{d(v, u)}, \quad (v, u) \in E. \quad (1)$$

Spatial encoder (per frame). GraphSAGE [7] updates every node’s representation by aggregating its neighbourhood. With edge weights, the neighbourhood mean becomes a distance-weighted average, and the update concatenates each node’s own (self) representation with that weighted-neighbour summary before a shared linear projection and ReLU:

$$\mathbf{h}_v^{(k)} = \sigma \left(\mathbf{W}^{(k)} \left[\mathbf{h}_v^{(k-1)} \parallel \frac{\sum_{u \in \mathcal{N}(v)} w_{vu} \mathbf{h}_u^{(k-1)}}{\sum_{u \in \mathcal{N}(v)} w_{vu}} \right] \right) \quad (2)$$

where $\mathcal{N}(v)$ denotes the k -NN neighbours of v , $\parallel$ is concatenation, and σ is ReLU. The same layer (with shared weights $\mathbf{W}^{(k)}$, output width `hidden_g`) is applied to all 256 nodes simultaneously and stacked $K \in \{2, 3, 4\}$ times (`n_sage_layers`); each additional layer extends a node’s effective receptive field by one more k -NN hop. A final *mean readout* over all $N = 256$ nodes collapses the per-pixel representations into a single graph-level (frame) embedding $\mathbf{g}_\ell \in \mathbb{R}^{d_g}$, with $d_g = \text{hidden_g} \in \{64, 96, 128, 192, 256\}$ — the GraphSAGE

analogue of the ResNet’s d_e -dimensional frame embedding.

Temporal encoder. As in ResNet-LSTM, the L graph-level embeddings are normalised and stacked into $(\mathbf{g}_1, \dots, \mathbf{g}_L) \in \mathbb{R}^{L \times d_g}$ and processed by a single-
280 or two-layer LSTM whose final hidden state feeds an identical MLP head producing $\hat{y}$. The two architectures are thus directly comparable: they differ *only* in how the per-frame spatial embedding is produced — convolution-and-downsampling over a regular grid (ResNet) versus weighted message-passing over a learned k -NN pixel graph at full 256-node resolution, with no spatial
285 downsampling (GraphSAGE) — while sharing the same temporal model, output head, training protocol, and data pipeline.

3.3.4. FlatMLP: temporal-only ablation

To isolate the contribution of *spatial* encoding from that of temporal modelling, FlatMLP discards all pixel-level spatial structure. For every frame $X_\ell \in$
290 $\mathbb{R}^{C \times P \times P}$, a global spatial average pool collapses the $P \times P$ grid to its per-channel mean, $\bar{\mathbf{x}}_\ell \in \mathbb{R}^C$ — i.e. each frame is reduced to a 16-dimensional “patch-mean spectral signature” with no notion of pixel location. The $L = 24$ resulting vectors are concatenated into a single flat input $\bar{\mathbf{x}} \in \mathbb{R}^{L \times C} = \mathbb{R}^{384}$ (24×16). This vector is passed through a fully-connected feed-forward net-
295 work of `n_layers` blocks, each consisting of `Linear` $\rightarrow$ `LayerNorm` $\rightarrow$ `ReLU` $\rightarrow$ `Dropout` (`hidden_dim` units), followed by a final `Linear` layer that outputs the scalar $\hat{y}$ directly — every unit in one layer is connected to every unit in the next (a standard dense MLP), with no recurrence, no convolution, and no graph structure. Because FlatMLP receives exactly the same $L \times C \times P \times P$ input
300 tensors (via `PatchSeqDataset`) and is trained with the same protocol, its skill gap relative to ResNet-LSTM and GraphSAGE-LSTM directly quantifies how much of their forecasting skill stems from *spatial* encoding of the satellite patch, as opposed to simple per-frame statistics processed over time.

3.4. Hyperparameter optimisation

305 For the three deep-learning models (ResNet-LSTM, GraphSAGE-LSTM, and FlatMLP) we use Optuna [9] with the TPE sampler and Median pruner. The study budget is 50 trials per configuration (site $\times$ horizon $\times$ seed) with $n_{\text{jobs}} = 2$; an extended run with 100 trials is being evaluated to confirm convergence.

310 The ResNet-LSTM and GraphSAGE-LSTM search spaces (Table 3) include `n_lstm_layers` $\in \{1, 2\}$ to allow stacked recurrent processing, and a categorical L1 regularisation coefficient (`l1_reg`) to penalise large weights in high-capacity configurations; both are new relative to the initial baseline study. For **ResNet-LSTM** specifically, the embedding dimension (`emb_dim`) and temporal hidden
315 size (`hidden_t`) are extended to 256 to increase capacity. For **GraphSAGE-LSTM**, the number of nearest neighbours ($k \in \{4, 8, 12, 16\}$) used to build the weighted pixel graph is additionally tunable, allowing the study to determine the optimal spatial connectivity for each site-horizon combination. For **FlatMLP**, the study searches over MLP depth, hidden dimension, dropout, learning rate,
320 and weight decay (50 trials).

The Optuna study identifies the best trial on validation RMSE_{day} ; the winning configuration is then retrained for up to 30 epochs with patience 8 and evaluated on the held-out test set.

3.5. Training protocol

325 All deep-learning models are trained with the Adam optimiser minimising the MSE loss on normalised targets ($y_{\text{norm}} = (y - \bar{y})/s_y$, with $\bar{y}$ and s_y estimated from the training set). Training uses early stopping with patience 6 (trials) or 8 (final retraining). Automatic Mixed Precision (AMP) is enabled when a GPU is available. Each experiment is replicated with four seeds $\{42, 1, 7, 13\}$ to report
330 mean $\pm$ std statistics.

3.6. Uncertainty quantification: Split Conformal Prediction

All point-forecast models are used as the backbone for a three-stage uncertainty quantification (UQ) pipeline whose long-term goal is to decompose total

Table 3: Optuna hyperparameter search spaces. L1 coefficient and LSTM depth are new relative to the initial baseline study. The `k_neighbors` parameter determines the weighted k -NN graph built per trial for GraphSAGE-LSTM.

Hyperparameter	ResNet-LSTM	GraphSAGE-LSTM
<code>base / hidden_g</code>	{16, 24, 32, 48}	{64, 96, 128, 192, 256}
<code>emb_dim / n_sage_layers</code>	{64, 128, 192, 256}	{2, 3, 4}
<code>hidden_t (LSTM)</code>	{64, 128, 192, 256}	{64, 96, 128, 192, 256}
<code>n_lstm_layers</code>	{1, 2}	{1, 2}
<code>k_neighbors</code>	—	{4, 8, 12, 16}
<code>l1_reg</code>	{0, 10^{-5} , 10^{-4} , 10^{-3} }	{0, 10^{-5} , 10^{-4} , 10^{-3} }
<code>dropout</code>	{0, 0.1, 0.2, 0.3}	{0, 0.1, 0.2}
<code>lr</code>	$[3 \times 10^{-4}, 3 \times 10^{-3}]$	$[5 \times 10^{-5}, 2 \times 10^{-3}]$
<code>weight_decay</code>	$[10^{-6}, 10^{-3}]$	$[10^{-6}, 10^{-3}]$
<code>batch_size</code>	{16, 32, 64}	{8, 16, 32}

forecast uncertainty into its *aleatoric* component (irreducible cloud stochasticity) and its *epistemic* component (uncertainty due to limited data and model capacity).

Stage 1 — Split Conformal Prediction (implemented, [10, 11]).

We apply Inductive Conformal Prediction as a distribution-free, finite-sample calibration method. A non-conformity score is computed over the validation set:

$$s_i = |y_i - \hat{y}_i| \quad [\text{W/m}^2]. \quad (3)$$

The calibrated conformal quantile at miscoverage level α is

$$\hat{q} = \text{Quantile}\left(\{s_i\}_{i=1}^{n_{\text{cal}}}, \frac{\lceil (n_{\text{cal}} + 1)(1 - \alpha) \rceil}{n_{\text{cal}}}\right), \quad (4)$$

and the prediction interval for a new test point is

$$\mathcal{C}_{1-\alpha}(x) = [\hat{y}(t + H) - \hat{q}, \hat{y}(t + H) + \hat{q}]. \quad (5)$$

Under exchangeability, this guarantees marginal coverage $\mathbb{P}(Y \in \mathcal{C}_{1-\alpha}) \geq 1 - \alpha$ exactly in finite samples [11]. Because $\hat{q}$ is a global constant, the interval has

345 fixed width $2\hat{q}$ and does not adapt to local prediction difficulty. We evaluate empirical coverage at $\alpha \in \{0.05, 0.10, 0.20\}$ (intervals of 95%, 90%, and 80%) reporting both overall and daytime-only metrics.

Stage 2 — Conformalized Quantile Regression (planned, [12]). To obtain intervals whose width adapts to local uncertainty, we will train quantile-
350 regression heads on the DL encoders and apply the CQR correction. This is expected to reduce unnecessary interval width during low-variability periods (clear-sky mornings) without sacrificing coverage.

Stage 3 — Bayesian uncertainty decomposition (planned, [13, 34]). Gaussian variance heads (NLL loss) will model aleatoric uncertainty, while
355 Stochastic Gradient Langevin Dynamics (SGLD, [34]) will sample the posterior weight distribution to quantify epistemic uncertainty. SGLD injects calibrated Langevin noise into gradient updates, enabling approximate Bayesian inference without the computational overhead of full MCMC: running the chain for T epochs and collecting snapshots every Δ epochs yields T/Δ approximate poste-
360 rior samples, whose ensemble mean serves as the point forecast and whose standard deviation quantifies model-knowledge uncertainty. Together, the aleatoric head and SGLD ensemble will decompose total forecast uncertainty into its irreducible (cloud stochasticity) and reducible (model-knowledge) components [13], supporting risk-stratified dispatch decisions for solar-integrated power systems.

365 3.7. Evaluation metrics

Following standard practice in solar forecasting [35], we report:

Daytime RMSE (RMSE_{day} , W/m^2): RMSE computed only over test samples with observed $\text{GHI} \geq 20 \text{ W/m}^2$, excluding night-time samples where all models trivially predict zero.

370 **Daytime skill score** ($\text{Skill}_{\text{day}}$): normalised improvement over persistence,

$$\text{Skill}_{\text{day}} = 1 - \frac{\text{RMSE}_{\text{day}}(\text{model})}{\text{RMSE}_{\text{day}}(\text{persistence})} \quad (6)$$

A score of 0 indicates parity with persistence; positive values indicate improvement. We also report MAE_{day} for completeness.

Table 4: Persistence test-set RMSE_{day} (W/m^2). All deep-learning models are compared against these values.

Site	1h	3h	6h
El Paso	204.4	411.0	565.3
Uniandes	293.9	405.0	470.9

4. Results

4.1. Persistence reference

375 Table 4 reports the persistence RMSE_{day} at each site and horizon, which defines the denominator of $\text{Skill}_{\text{day}}$ for all models.

The 1-hour persistence error at El Paso ($204.4 \text{ W}/\text{m}^2$) is notably lower than at Uniandes ($293.9 \text{ W}/\text{m}^2$), reflecting El Paso’s more stable clear-sky regime where short-horizon irradiance changes are small. At 6 hours, both sites have
380 comparable absolute difficulty (565.3 vs. $470.9 \text{ W}/\text{m}^2$).

4.2. Main results

Table 5 reports RMSE_{day} and $\text{Skill}_{\text{day}}$ for all models on the test set. Deep-learning results are mean $\pm$ std over four seeds.

4.3. Horizon-by-horizon analysis

385 *1-hour horizon..* Both deep-learning models fail to beat persistence at El Paso: ResNet-LSTM ($\text{Skill}_{\text{day}} = -0.167 \pm 0.110$) and GraphSAGE-LSTM (-0.197 ± 0.040) show negative skill, meaning they produce higher RMSE than simply repeating the last observation. At Uniandes the picture is more favourable: ResNet-LSTM achieves modest positive skill ($+0.161 \pm 0.014$), narrowly ahead
390 of GraphSAGE-LSTM ($+0.147 \pm 0.017$). The key difference between sites is the persistence reference: El Paso’s 1-hour persistence RMSE ($204.4 \text{ W}/\text{m}^2$) is comparatively low because irradiance changes slowly under predominantly clear-sky conditions, leaving little room for improvement from data-driven methods.

Table 5: Test-set results. RMSE_{day} in W/m^2 . $\text{Skill}_{\text{day}}$ relative to persistence (higher is better). DL models: mean $\pm$ std over 4 seeds. Best $\text{Skill}_{\text{day}}$ per site-horizon highlighted in **bold**.

Model	Metric	El Paso (César)			Uniandes (Bogotá)		
		1h	3h	6h	1h	3h	6h
Persistence	RMSE_{day}	204.4	411.0	565.3	293.9	405.0	470.9
	$\text{Skill}_{\text{day}}$	0.000	0.000	0.000	0.000	0.000	0.000
ResNet-LSTM	RMSE_{day}	238.6 ± 22.4	238.5 ± 42.7	218.4 ± 13.2	246.7 ± 4.1	255.7 ± 4.8	297.9 ± 8.8
	$\text{Skill}_{\text{day}}$	-0.167 ± 0.110	$+0.420 \pm 0.104$	$+0.614 \pm 0.023$	$+0.161 \pm 0.014$	$+0.369 \pm 0.012$	$+0.368 \pm 0.019$
GraphSAGE-LSTM	RMSE_{day}	244.7 ± 8.1	238.9 ± 15.9	206.1 ± 7.4	250.8 ± 5.1	252.5 ± 1.5	274.4 ± 3.9
	$\text{Skill}_{\text{day}}$	-0.197 ± 0.040	$+0.419 \pm 0.039$	$+0.636 \pm 0.013$	$+0.147 \pm 0.017$	$+0.376 \pm 0.004$	$+0.417 \pm 0.008$
FlatMLP (Optuna)	RMSE_{day}	—	—	—	—	—	—
	$\text{Skill}_{\text{day}}$	—	—	—	—	—	—

FlatMLP (Optuna) results are pending completion of the four-seed sweep (Section 5, Limitations) and will be populated prior to submission; this row quantifies what fraction of deep-learning skill arises from spatial encoding versus temporal dynamics alone.

Uniandes’ 1-hour persistence RMSE ($293.9 \text{ W}/\text{m}^2$) is higher due to the frequent
 395 cloud passages, which create more structure for the models to exploit.

3-hour horizon.. Both deep-learning models outperform persistence at El Paso by approximately 42% ($\text{Skill}_{\text{day}} \approx 0.42$ for both architectures, with GraphSAGE-LSTM marginally ahead at $+0.419 \pm 0.039$). The large standard deviation for ResNet-LSTM (± 0.090) reflects seed-to-seed variability, with some seeds achieving
 400 ing near-GraphSAGE performance and others considerably below.

At Uniandes, GraphSAGE-LSTM attains the best skill at this horizon ($+0.376 \pm 0.004$), narrowly ahead of ResNet-LSTM ($+0.369 \pm 0.012$); both are similar to each other and slightly below their El Paso counterparts despite the higher absolute persistence error at this site.

405 *6-hour horizon..* This is the most favourable horizon for the deep-learning models. **GraphSAGE-LSTM achieves the best overall result:** $\text{RMSE}_{\text{day}} = 206.1 \pm 7.4 \text{ W}/\text{m}^2$ and $\text{Skill}_{\text{day}} = 0.636 \pm 0.013$ at El Paso, reducing the persistence RMSE by 63.6%. ResNet-LSTM is close (0.614 ± 0.023) but consistently below GraphSAGE across all seeds.

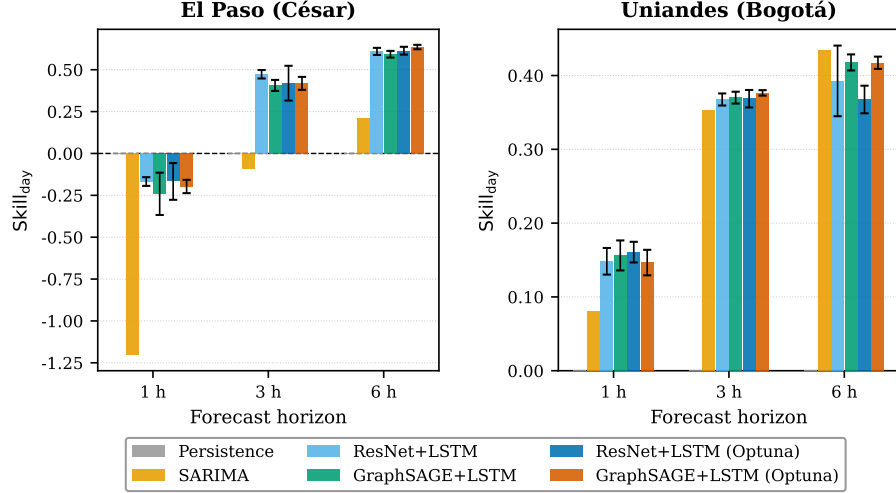


Figure 1: Daytime skill score ($\text{Skill}_{\text{day}}$) for the deep-learning models and persistence at both sites across three forecast horizons. Bars show mean over four seeds; error bars indicate ± 1 standard deviation. Persistence has no seed variability. GraphSAGE-LSTM (Optuna) achieves the highest skill among the deep-learning models at the 6-hour horizon at both sites (0.636 at El Paso; 0.417 at Uniandes).

At Uniandes, GraphSAGE-LSTM again leads (0.417 ± 0.008 vs. 0.368 ± 0.019 for ResNet-LSTM), with a wider margin than at El Paso ($+0.049$ vs. $+0.022$ percentage points of skill). However, the *absolute* skill at Uniandes remains markedly lower than at El Paso for both architectures — the gap we analyse in Section 5.

4.4. Statistical robustness

The standard deviation across seeds provides a measure of training stability. GraphSAGE-LSTM consistently shows lower seed-to-seed variability than ResNet-LSTM (e.g., El Paso 6h: ± 0.013 vs. ± 0.023), suggesting that the graph structure regularises training and reduces sensitivity to weight initialisation. The most unstable model is ResNet-LSTM at El Paso 3h (± 0.090), which may reflect the interaction between the high-irradiance regime and the CNN’s sensitivity to specific initialisation-driven features.

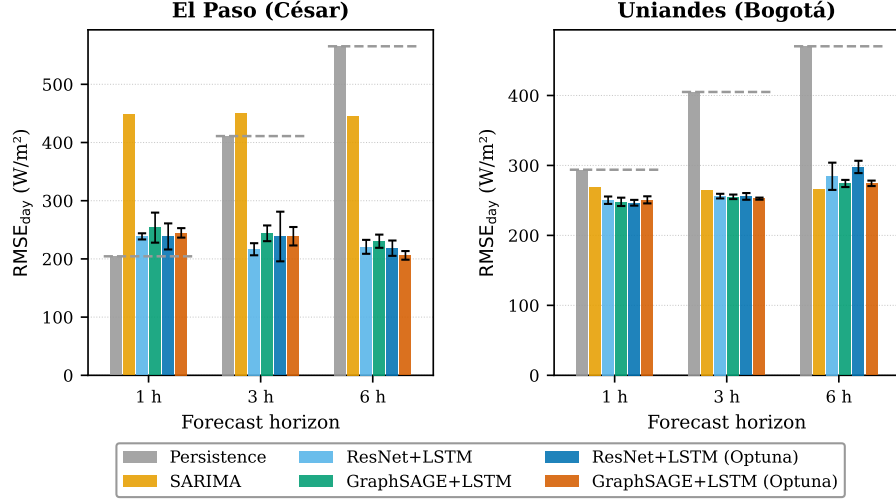


Figure 2: Daytime RMSE (RMSE_{day} , W/m^2) for the deep-learning models and persistence. Dashed horizontal lines show the persistence reference for each horizon group. Both deep-learning models substantially reduce RMSE relative to persistence at 3h and 6h, with GraphSAGE-LSTM achieving the lowest RMSE at the 6-hour horizon at both sites.

4.5. Conformal prediction intervals

Table 6 will report the Split CP calibration results for the best-performing model at each site-horizon combination, evaluated at nominal coverage levels 80%, 90%, and 95%. The conformal quantile $\hat{q}$ (W/m^2) sets the half-width of the symmetric prediction interval; daytime empirical coverage côv_{day} should meet or exceed the nominal level to satisfy the marginal coverage guarantee of Eq. (4).

The backbone for El Paso is GraphSAGE-LSTM (best overall model at 3h and 6h); for Uniandes the best deep-learning backbone per horizon is used (ResNet-LSTM at 1h, GraphSAGE-LSTM at 3h and 6h), consistent with Table 5.

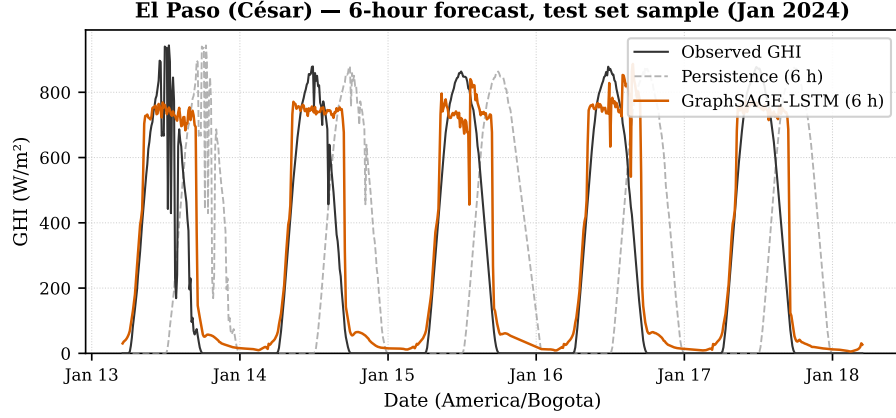


Figure 3: Five-day excerpt of the 6-hour GHI forecast from the best model (GraphSAGE-LSTM Optuna, seed 42) at El Paso in January 2024. Persistence (dashed) simply repeats the GHI observed 6 hours prior, while GraphSAGE-LSTM anticipates the diurnal rise and cloud events. Timestamps in local time (America/Bogotá, UTC−5).

5. Discussion

5.1. Why does GraphSAGE outperform ResNet at longer horizons?

At the 6-hour horizon for El Paso, GraphSAGE-LSTM achieves a 2.2-percentage-point skill improvement over ResNet-LSTM (0.636 vs. 0.614). We hypothesise two mechanisms:

Irregular spatial structure. At longer lead times, the relevant cloud signal migrates from the centre of the patch (where it directly causes GHI reduction) toward the periphery, where incoming cloud advection is reflected in the satellite channels. CNN filters, being translation-equivariant, apply the same spatial template across all positions; GraphSAGE, by contrast, learns separate aggregation rules that can weight distant pixels differently depending on their graph neighbourhood. This flexibility may be especially valuable for detecting cloud-edge gradients that precede GHI events.

Reduced redundancy. The ResNet applies multiple stacked convolutions that effectively redundantly process the spatial information across layers; the GraphSAGE message-passing is applied once per hop and is less prone to over-

Table 6: Split Conformal Prediction results. $\hat{q}$: calibrated half-width (W/m^2); côv_{day} : empirical daytime coverage on the test set. Three nominal levels $1 - \alpha \in \{80\%, 90\%, 95\%\}$ are reported per site-horizon combination. Backbone: GraphSAGE-LSTM for El Paso; best-per-horizon deep-learning backbone for Uniandes (ResNet-LSTM at 1h, GraphSAGE-LSTM at 3h and 6h). *[Conformal calibration in progress; values to be populated prior to submission.]*

Level	H	El Paso			Uniandes		
		$\hat{q}$	côv_{day}	Width	$\hat{q}$	côv_{day}	Width
80%	3h	—	—	—	—	—	—
80%	6h	—	—	—	—	—	—
90%	3h	—	—	—	—	—	—
90%	6h	—	—	—	—	—	—
95%	3h	—	—	—	—	—	—
95%	6h	—	—	—	—	—	—

450 smoothing at the patch scale ($P = 16$). The lower seed variance of GraphSAGE (± 0.013 vs. ± 0.023 at El Paso 6h) is consistent with a more stable loss landscape.

5.2. Why is forecasting skill lower at Uniandes than at El Paso?

Across all deep-learning models and horizons, absolute skill is markedly lower at the Uniandes mountain site than at El Paso (e.g., GraphSAGE-LSTM at 6h: 455 $\text{Skill}_{\text{day}} = 0.417$ vs. 0.636). We attribute this gap to two related properties of the Bogotá climate:

Diurnal regularity narrows the room for improvement. At 4.6°N , the solar declination varies little across the year ($\pm 4.6^\circ$), producing a highly repeatable clear-sky envelope. Cloud formation in Bogotá is predominantly 460 triggered by orographic lifting in the early afternoon, generating a strong, regular diurnal cycle. Much of the predictable structure in the Uniandes GHI series is therefore already encoded in its own recent history and time-of-day, leaving comparatively less *additional* information for a 4-hour satellite patch sequence to contribute — which compresses the achievable skill ceiling for every model,

465 spatially-aware or not.

Satellite signal saturation at longer horizons. At 6 hours, the satellite patch observed at t may carry little predictive information about the ground GHI at $t + 6$ h: the cloud system that will eventually affect the site has often not yet entered the 16×16 -pixel field of view, particularly at a site where convection
470 forms locally through orographic lifting rather than advecting in from a stable upstream direction. This limits how much spatial encoding — regardless of architecture — can contribute at the longest horizon.

This result suggests that *site-specific climatological structure should inform both model design and expectation-setting*: in climates with strong, regular di-
475 urnal cloud patterns and locally-forming convection, the marginal benefit of spatial deep learning over simpler temporal signals is smaller, and reported skill scores should always be interpreted relative to the local persistence baseline rather than compared at face value across sites.

5.3. Why is the 1-hour horizon intractable at El Paso?

480 The negative skill at 1-hour El Paso (all DL models, $\text{Skill}_{\text{day}} \approx -0.17$ to -0.20) can be explained by the *low persistence error*: at 204.4 W/m^2 , persistence is already very accurate because the predominantly clear-sky regime means irradiance rarely drops by more than a small fraction over 10 minutes. The DL models, trained to minimise MSE over the full distribution (includ-
485 ing the occasional cloud event), learn a slightly smoothed version of the target trajectory that incurs higher error on the flat clear-sky majority.

A potential remedy is a loss function that differentially weights cloud events relative to the clear-sky envelope (e.g., scaling the per-sample MSE by the departure of observed GHI from the clear-sky reference), or dedicated cloud-event
490 detection as a preprocessing step.

5.4. Practical implications

- For **El Paso-type sites** (high irradiance, moderate variability), GraphSAGE-LSTM with Optuna-tuned hyperparameters is the recommended model at

3h and 6h horizons, with a 60+% reduction in RMSE over persistence. At
 495 1h, the system should fall back to persistence or a light-weight nowcasting
 model.

- For **Uniandes-type sites** (tropical mountain, high convective variability), even the best models achieve markedly lower skill than at El Paso (Skill_{day} ≤ 0.42 vs. 0.64 at 6h); operators should calibrate reserve margins
 500 to the local persistence baseline rather than assume that skill levels transfer across climates. GraphSAGE-LSTM remains the most consistent top performer at 3h and 6h, but gains over persistence are smaller and more sensitive to careful hyperparameter tuning.
- The high seed variance of ResNet-LSTM at El Paso 3h (±0.090) suggests
 505 that multiple retraining runs should be averaged for production deployment, or that GraphSAGE-LSTM should be preferred for its higher stability.

5.5. Uncertainty quantification: from point forecasts to calibrated intervals

Grid operators require probabilistic outputs — prediction intervals that are
 510 well-calibrated across conditions of varying cloud cover — not just point estimates. Quantifying and ultimately decomposing forecast uncertainty is the central long-term goal of this work.

Stage 1: Split Conformal Prediction (implemented). Because all models produce only point forecasts, we apply Split CP [10, 11] as a distribution-
 515 free post-hoc layer. The calibrated interval $[\hat{y} \pm \hat{q}]$ (Section 3.6) carries a finite-sample marginal coverage guarantee regardless of the model residual distribution, which is particularly valuable for tropical sites where the error distribution is heavy-tailed due to cloud passages. A key limitation is that $\hat{q}$ is globally constant: the interval does not narrow during predictable clear-sky mornings nor
 520 widen during uncertain afternoon convection at Uniandes.

Stage 2: Conformalized Quantile Regression (planned). CQR [12] replaces the symmetric residual score with a quantile-regression score that adapts

the interval width to local difficulty. This is expected to reduce average interval width substantially at El Paso (predominantly clear-sky) while maintaining coverage.

Stage 3: Aleatoric/epistemic decomposition (planned). Gaussian variance heads (NLL loss) and MC Dropout will provide separate estimates of irreducible measurement noise (aleatoric) and model-knowledge uncertainty (epistemic). Under the formulation of [13], SGLD [34] can further sample the posterior weight distribution for a fully Bayesian treatment.

5.6. Implications for solar-integrated power system operation

The results presented here connect directly to operational decision-making in solar-integrated power systems.

Situational awareness under cloud-induced uncertainty. GraphSAGE-LSTM, by learning spatially-aware representations of incoming cloud systems from satellite imagery, provides operators with earlier detection of impending irradiance ramps — the primary source of generation-side uncertainty in PV-heavy grids. The 63.6% RMSE reduction over persistence at 6h for El Paso translates to significantly tighter dispatch intervals and reduced reserve requirements.

Calibrated prediction intervals for risk-aware dispatch. The Split CP layer converts any point forecast into an interval with a finite-sample coverage guarantee. For a system operator, this means that reserve scheduling can be tied to an explicit coverage target (e.g., 90% intervals) rather than to ad hoc safety margins. The global constant width $2\hat{q}$ is an acknowledged limitation, but even symmetric intervals provide stronger risk-management guarantees than unquantified point forecasts.

Climate- and horizon-adaptive model selection. The marked difference in absolute skill between El Paso (up to $\text{Skill}_{\text{day}} = 0.636$ at 6h) and Unian-des (up to 0.417) has a direct operational implication: forecasting pipelines should be benchmarked and calibrated locally rather than assuming that skill levels transfer across climates. Within a site, the best architecture also depends

on the horizon — persistence is hard to beat at 1h in the clear-sky-dominated El Paso regime, while GraphSAGE-LSTM is the most consistent top performer at 3h and 6h at both sites. A horizon-aware portfolio strategy — a light-weight nowcasting fallback at very short horizons, spatial deep learning at multi-hour horizons — could optimise the accuracy-cost tradeoff for operational deployment.

5.7. Limitations

Geographic scope. Both sites are located in Colombia and represent two tropical climate sub-types. Generalisation to other tropical or subtropical regions requires additional validation.

Flat MLP ablation in progress. The FlatMLP experiments are currently completing their Optuna runs (24/24 configurations being evaluated). Final results will be incorporated to quantify what fraction of deep-learning skill arises from spatial encoding versus temporal dynamics alone.

Single-step output. Current models produce a single-step forecast at horizon H . Simultaneous multi-step output (forecasts for all three horizons from a single forward pass) could improve consistency and reduce inference cost in real-time operational forecasting pipelines.

6. Conclusion

We presented a graph neural network framework for satellite-based solar irradiance forecasting that addresses the situational-awareness needs of solar-integrated power system operators in tropical climates. The key contributions are: (i) a weighted k -NN GraphSAGE-LSTM architecture that explicitly models spatial cloud structure from GOES-16 patches; (ii) a rigorous comparative evaluation against a ResNet-LSTM baseline, a flat-MLP spatial-ablation baseline, and persistence at two Colombian sites with contrasting climatic regimes; and (iii) Split Conformal Prediction as a distribution-free post-hoc UQ layer that transforms point forecasts into coverage-guaranteed prediction intervals usable in risk-aware operational decisions.

GraphSAGE-LSTM achieves the best skill among the deep-learning models at the 6-hour horizon at *both* sites (El Paso: $\text{Skill}_{\text{day}} = 0.636$ vs. 0.614 for ResNet-LSTM; Uniandes: 0.417 vs. 0.368), with lower seed variance throughout, confirming that distance-weighted neighbourhood aggregation on the pixel graph provides a more stable and expressive spatial representation than fixed convolutional filters.

At the **tropical mountain site** (Uniandes), however, the absolute skill of every model is markedly lower than at El Paso even at the most favourable horizon ($\text{Skill}_{\text{day}} \leq 0.42$ vs. 0.64 at 6h), revealing that the strong diurnal autocorrelation of the Bogotá clear-sky cycle and the locally-driven nature of orographic convection limit the marginal benefit of satellite spatial features at this site. This finding has a direct operational implication: model selection and skill expectations should be climate-aware, and forecasting pipelines should always be benchmarked and calibrated against the local persistence reference rather than assumed to transfer across sites.

At the **1-hour horizon**, no model consistently beats persistence at El Paso, consistent with the low persistence error (204.4 W/m^2) in a predominantly clear-sky regime. Modest positive skill (≈ 0.15) is achieved at Uniandes, where frequent cloud passages create exploitable structure.

Future work follows a staged uncertainty-quantification roadmap toward fully probabilistic, operationally-deployable solar forecasts. Split Conformal Prediction has been implemented as Stage 1; Conformalized Quantile Regression [12] will yield adaptive-width intervals that contract during predictable clear-sky conditions and expand during uncertain cloud events (Stage 2), enabling tighter reserve scheduling. Stage 3 will achieve full Bayesian decomposition of forecast uncertainty: Gaussian variance heads will capture aleatoric uncertainty (irreducible cloud stochasticity), while Stochastic Gradient Langevin Dynamics (SGLD, [34]) will sample the posterior weight distribution to provide epistemic uncertainty estimates [13], completing the transition from point forecasts to risk-stratified probabilistic outputs for solar-integrated power system operation. Additional planned work includes the FlatMLP spatial-ablation analysis, multi-

step joint-horizon outputs, and extension to further Latin American sites to build a broader tropical solar forecasting benchmark.

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